Differential effects of reproductive factors on the risk of pre- and postmenopausal breast cancer. Results from a large cohort of French women

The aim of this study was to obtain a better understanding of the role of hormonal factors in breast cancer risk and to determine whether the effect of reproductive events differs according to age at diagnosis. It analysed the effect of age at menarche, age at first full-term pregnancy, number of full-term pregnancies and number of spontaneous abortions both on the overall risk of breast cancer and on its pre- or postmenopausal onset, using the data on 1718 breast cancer cases, obtained from a large sample of around 100 000 French women participating in the E3N cohort study. The results provide further evidence that the overall risk of breast cancer increases with decreasing age at menarche, increasing age at first pregnancy and low parity. No overall effect of spontaneous abortions was observed. The effect of these reproductive factors differed according to menopausal status. Age at menarche had an effect on premenopausal breast cancer risk, with a decrease in risk with increasing age of 7% per year ($P<0.05$). Compared to those who had their first menstrual periods at 11 or before, women experiencing menarche at 15 or after had an RR of 0.66 (95% CI 0.45–0.97) in the premenopausal group. Age at first full-term pregnancy had an effect on both pre- and postmenopausal breast cancer risk, with significant tests showing increasing risk per year of increasing age ($P=0.001$ and $P<0.05$ respectively). A first full-term pregnancy above age 30 conveyed a risk of 1.63 (95% CI 1.12–2.38) and 1.35 (95% CI 1.02–1.78) in the pre- and postmenopausal groups respectively. A protective effect of high parity was observed only for postmenopausal breast cancer risk (P for trend test =0.001), with point estimates of 0.79 (95% CI 0.60–1.04), 0.69 (95% CI 0.54–0.88), 0.66 (95% CI 0.51–0.85) and 0.64 (95% CI 0.48–0.86) associated to a one, two, three and four or more full-term pregnancies. A history of spontaneous abortion had no significant effect on the risk of breast cancer diagnosed before or after menopause. Our results suggest that reproductive events have complex effects on the risk of breast cancer. British Journal of Cancer (2002) 86, 723–727. DOI: 10.1038/sj/bjc/6600124 www.bjcancer.com © 2002 Cancer Research UK